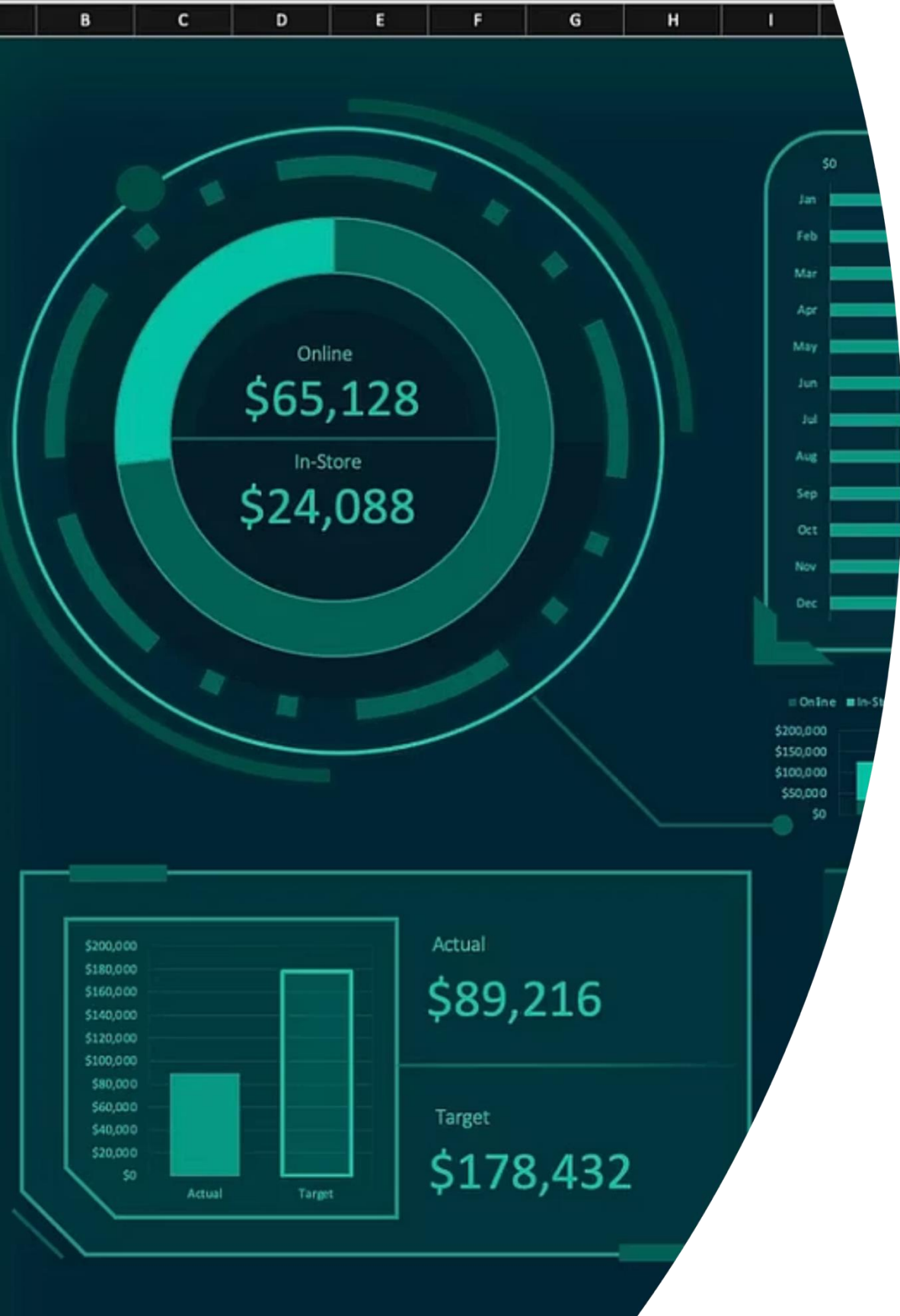




Credit Card Portfolio Analytics

Banking & Financial Services Sector

G-17

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Context & Problem Statement

Core Question

How can we optimise portfolio profitability whilst effectively managing risk using granular customer-level analytics?

Strategic Objectives

- Identify high-value customer segments
- Quantify risk exposure patterns
- Uncover revenue growth opportunities
- Enable data-driven portfolio decisions

Decision Context

Credit card portfolios balance three critical dimensions: revenue generation through interest and fees, active customer engagement, and controlled credit risk exposure. Portfolio managers require actionable intelligence to optimise this balance.

Data Engineering Pipeline

Source Dataset

Raw Data containing 8,950 customer records across 18 behavioural variables including balance, credit limit, purchases, cash advance, payments, and tenure metrics.

Data Quality Issues

- Missing values identified in minimum payments field
- Extreme balance-to-limit ratios requiring investigation
- Zero balance accounts affecting ratio calculations

Comprehensive data cleansing protocols were implemented prior to KPI derivation to ensure analytical integrity.

KPI & Metrics Framework

Financial Strength

- Total Balance: **\$14.0M**
- Payment Ratio (Median): **1.39**
- Average Credit Limit: **\$4,494**
- Portfolio-Wide Utilization: **38.9%**

Risk Exposure

- High Utilization: **20.25%**
- High Cash Dependency: **36.26%**
- 48.29% Use Cash Advances

Behavioral Stability

- Median Tenure: **12 months**
- Majority Represent Mid-Tier Customers
- Payment Consistency Varies by Segment

These KPIs enable **dynamic portfolio steering**, not just retrospective reporting.

Insight 1: Hidden Revenue Opportunity

49% Underutilize Credit

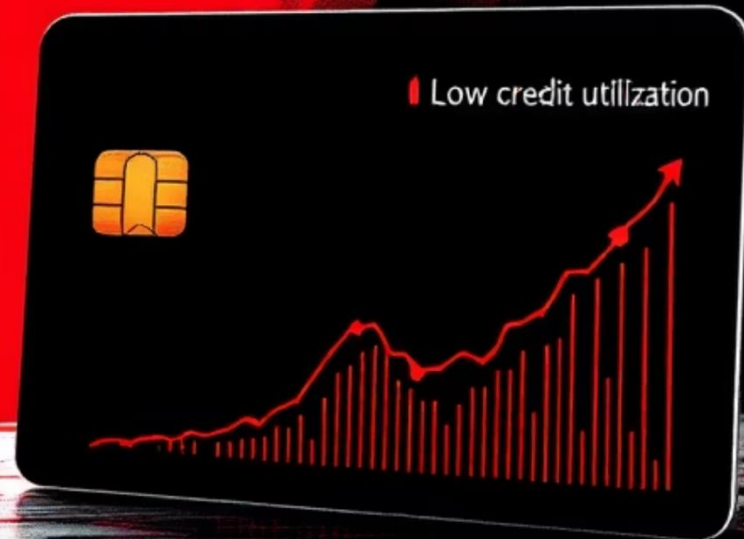
Nearly half the portfolio uses less than 30% of their available credit limit.

Low-Risk Expansion

These customers represent scalable revenue potential with minimal incremental risk exposure.

Strategic Action

Selective credit line increases can drive yield without compromising portfolio quality.



Insight 2: Hidden Revenue Opportunity



High Leverage Concentration

20% of customers are highly leveraged with utilisation exceeding 80%, serving as an early stress indicator requiring enhanced monitoring protocols and potential intervention strategies

Portfolio Liquidity Position

Total payments of **£15.5M exceed outstanding balance** of £14.0M, suggesting overall portfolio liquidity. However, risk remains concentrated within specific customer segments, necessitating granular analysis and differentiated strategies.

Liquidity Dependence Signal

Almost **48% utilise cash advances**, with 36% showing high dependency patterns. This concentration indicates liquidity constraints and represents a material risk segment requiring targeted management.

Key Insights (EDA)

Revenue Depends on Few Customers

- **Persistent Debt Cycles:** Only **15.4% of the customer base** pays their statement in full each month. This indicates that the vast majority are "revolvers," providing a stable interest-based revenue stream but also highlighting a segment sensitive to interest rate changes.
- **One-Off vs. Installment Behavior:** Large, single-ticket purchases (\$592 avg) significantly outweigh structured installment purchases (\$411 avg). Users are using their cards for immediate "big buys" rather than planned budget spreading.
- **Liquidity over Shopping:** A significant group of users relies on the card primarily for **Cash Advances (~\$978 avg)** rather than retail purchases. For these users, the card is a liquidity tool (emergency cash) rather than a lifestyle shopping tool.
- **The "Sleeper" Majority:** Over **50% of your database** (4,522 users) falls into a "Low Activity" category, with average balances under \$500 and minimal purchase frequency. This represents a massive untapped opportunity for re-engagement.
- **Frequency as a Growth Lever:** There is a strong correlation (0.69) between the number of transactions and total purchase volume. Frequent small-ticket shoppers eventually transition into high-value customers more reliably than infrequent big-spenders.
- **Credit Cushion:** The average credit limit is **\$4,494**, while the average balance is **\$1,564**. Most customers have significant "open-to-buy" space (approx. 65% available), suggesting that the barrier to spending is not the credit limit, but customer engagement.

Advanced Analysis: Behavioral Segmentation

Analysis: Applied K-Means Clustering to 17 behavioral variables to move beyond demographics into "intent-based" grouping.

Segment 1: The VIPs (<1% of base)

Profile: Extreme high-volume transactors (Avg. \$22k purchases).
Behavior: High credit limits (\$15k) and high payment-to-spend ratios. These are your most valuable, lowest-risk brand advocates.

Segment 2: Active Growth (17% of base)

Profile: Daily users with a near-perfect purchase frequency (0.94).
Behavior: Healthy utilization of healthy limits (\$7.1k). This is the "Engine Room" of your daily transaction revenue.

Segment 3: Cash-Dependent (32% of base)

Profile: High cash-advance usage (\$2.4k avg) vs. minimal shopping (<\$250).
Behavior: Likely using the card for liquidity or debt-cycling. This group represents the highest potential for future delinquency.

Segment 4: Prudent Sleepers (50.5% of base)

Profile: The "Silent Majority" with very low balances (\$484 avg).
Behavior: They possess the card but rarely use it. They are the primary target for activation and market-share growth.

New Understanding: The "Quality of Debt"

Volume vs. Velocity: This analysis revealed that a "High Balance" is not a single risk metric. A high balance in Segment 1 is "Good Debt" (high transaction volume), whereas a high balance in Segment 3 is "Risky Debt" (cash-seeking behavior).

Strategic Shift: By focusing on Purchase Frequency rather than just Balance, we can identify which customers are loyal shoppers and which are just using the card as an emergency loan.

Dashboard Walkthrough

1. Executive View: Strategic Oversight

High-level KPIs for leadership to monitor portfolio health and market share.

- **Portfolio Health Gauge:** A real-time ratio of Purchase Volume vs. Cash Advance Volume. A shift toward purchases indicates a healthier, lower-risk portfolio.
- **Customer Mix (Donut Chart):** Visual distribution of the 4 key segments. Highlights the 50.5% "Sleeper" population as a primary growth opportunity.
- **Revenue at Risk Trend:** A 12-month look at "Minimum Payment Only" users, flagging potential delinquency before it hits the balance sheet.
- **Capital Deployment Score:** Tracks total Credit Limit vs. Balance to identify if capital is being allocated to the most active users.

2. Operational View: Tactical Drill-Down

Granular data for Risk and Marketing teams to take immediate action.

- **Utilization Heatmap:** A grid cross-referencing Credit Limit Tiers with Tenure. Identifies "safe" long-term customers who are under-utilizing their cards.
- **Cash-Spiraling Alert System:** A filtered list of customers with high Cash Advance Frequency and declining Payment Ratios, prioritized for risk intervention.
- **Campaign Lead Generator:** A searchable table of CUST_IDs with high Purchase Frequency but low Credit Limits, earmarked for proactive Limit Increases.
- **Behavioral Deep-Dive:** Interactive filters to view spending patterns (One-off vs. Installments) by Tenure (6–12 months) to tailor loyalty offers.

Recommendations

Dormant Account Activation (The "Sleeper" Strategy)

- Target: The 4,522 customers (50.5% of the base) currently showing low utilization.
- Action: Launch a "First 3 transactions" incentive program (e.g., \$25 statement credit) to increase Purchase Frequency.
- Goal: Move these users from "emergency use" cards to "top-of-wallet" status.

Risk-Based "Cash-to-Purchase" Migration

- Target: The 2,867 customers (32% of base) with high cash-advance dependency (\$2.4k avg).
- Action: Offer targeted 3% cashback on essential retail categories (groceries/utilities) for 90 days.
- Goal: De-risk the portfolio by shifting usage from high-risk unsecured cash advances to safer, merchant-interchange-generating transactions.

Strategic Credit Line Increases (CLI)

- Target: The 1,523 "Active Growth" customers with high purchase frequency (0.94) and stable payments.
- Action: Proactively offer a 20% limit increase to those with a Full Payment Ratio > 20% (above the 15% average).
- Goal: Preemptively capture larger ticket "One-Off" purchases (\$592 avg) that would otherwise go to competitors.

Installment Conversion Program

- Target: High-balance "Revolving" customers who are not paying in full.
- Action: Implement an automated prompt in the mobile app to convert "One-Off" purchases over \$500 into a 6-month structured Installment Plan.
- Goal: Provide customers with a path to debt clearance while securing a predictable interest margin for the bank.

Impact & Value

The "So What?": Estimated Annual Impact

Financial Growth & Risk Mitigation

\$1.2M

Revenue Growth

in annual purchase volume

5%

Risk Reduction

decrease in high-risk unsecured exposure

20–30%

Marketing Efficiency

improvement in Campaign ROI

- **Revenue Growth: +\$1.2M in annual purchase volume.**
 - *The Logic:* Transitioning just **10%** of "Sleeper" customers to "Active" habits captures the **\$2,694 spend gap** per customer.
- **Risk Reduction: 5% decrease in high-risk unsecured exposure.**
 - *The Logic:* Proactive intervention in the "Cash-Dependent" segment reduces reliance on risky cash advances (currently **\$2.4k** average).
- **Marketing Efficiency: 20–30% improvement in Campaign ROI.**
 - *The Logic:* Eliminates "Mass-Blast" spending by targeting high-propensity segments (e.g., reserving VIP rewards for the top **1%**).

Slide 10 – Limitations & Next Steps

Data Gaps & Biases

Lack of Demographics: The absence of age, income, and occupation data prevents us from understanding the "Why" behind the "What." High cash-advance usage in a high-income bracket suggests a different risk profile than in a low-income bracket.

Static Snapshot: This analysis is a point-in-time view. We cannot distinguish between a "Sleeper" who is naturally low-activity and a previously "VIP" customer who is currently churning.

Next Steps (The Roadmap)



Time-Series Analysis

Incorporate the last 12 months of payment history to build a predictive Churn Model and identify seasonal spending peaks.



Credit Bureau Integration

Overlay external credit scores to see if our "Cash-Dependent" users are also seeking liquidity elsewhere, which would escalate their risk priority.



A/B Testing

Launch the "Sleeper Wake-Up" campaign on a controlled test group of 500 users. Measure the lift in transaction frequency over 60 days before a full portfolio-wide rollout.